

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.5: Rates of Reactions — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.5: Rates of Reactions
Driving Question	DRIVING QUESTION: Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is a Rate of Reaction? — Anchoring Phenomenon Launch	Two beakers containing identical amounts of sodium thiosulfate solution and dilute hydrochloric acid were placed over paper crosses. Beaker A was in cold water ($\approx 10^\circ\text{C}$ — like a cool Kericho morning) and Beaker B was in warm water ($\approx 50^\circ\text{C}$ — like a hot Mombasa afternoon). A milky-yellow solid (sulfur) formed in both beakers, but the cross under Beaker B disappeared in approximately 30–40 seconds, while the cross under Beaker A remained visible for several minutes.	The rate of a chemical reaction is a measure of how quickly reactants are used up or products are formed over time. In this experiment, the rate could be measured by timing how long it took for the cross to disappear. The two beakers had the same chemicals in the same amounts, yet their rates of reaction were dramatically different — something about the conditions of the reaction was changing the speed.	We cannot yet fully explain WHY the warm beaker reacted faster. Our initial particle models suggest that something at the particle level inside the warm beaker is different — perhaps the particles are moving differently or interacting more frequently. We have opened our Driving Question Board with questions about temperature, particle movement, and other possible factors. These questions will be investigated and answered across Lessons 2–8.
Lesson 2 What Is Rate of Reaction? — Measuring the Speed of Change	Learners repeated the sodium thiosulfate and HCl cross-disappearance experiment at room temperature ($\sim 25^\circ\text{C}$, typical Nairobi classroom conditions), recording three timed trials per group and pooling class results into a shared data table. They observed natural variation between groups' times and discussed sources of experimental error.	Learners formally defined rate of reaction as the change in a measurable quantity per unit time, and expressed it as $1/t$ (s^{-1}) for the cross-disappearance method. They compared three standard measurement methods (precipitate timing, gas volume collection, loss of mass) and connected each to a Kenyan real-life example (thiosulfate cross, Eno antacid fizz, burning mkaa on a jiko).	Learners constructed an initial particle model distinguishing Beaker A (cold, slow particles, slow rate) from Beaker B (warm, fast-moving particles, fast rate), labelling the rate values calculated from Lesson 1 data. They posted new questions on the Driving Question Board identifying temperature as the first confirmed variable — while noting that concentration, surface area, catalysts, and light remain unexplained, to be investigated in Lessons 3–7.

Lesson 3 Concentration and Collision Frequency: Why Does a More Concentrated Acid React Faster?	When we added dilute HCl to four flasks of sodium thiosulfate solution at different concentrations (holding temperature constant), the cross under the most concentrated flask disappeared in the shortest time, and the cross under the most dilute flask took the longest. When we calculated rate as 1/time and plotted a graph, the rate increased as concentration increased.	Increasing the concentration of a reactant increases the rate of reaction. This is because a higher concentration means more reactant particles are present in the same volume, so particles collide more frequently. More frequent effective collisions mean more product forms per second — a faster reaction rate. This is the same collision theory we used to explain temperature in Lesson 1, but today particle SPEED stayed the same while particle NUMBER per unit volume changed.	The disappearing-cross phenomenon from Lesson 1 can now be partially explained by concentration as well as temperature: if Beaker B had a higher concentration of thiosulfate AND a higher temperature, both factors would independently increase collision frequency and make the cross disappear faster. In Kenyan contexts: a higher concentration of HCl in the stomach speeds up protein digestion in githeri; a more concentrated lactic acid solution in uji ferments (sours) faster; and more acidic (concentrated H^+) rainfall corrodes iron roofing sheets in Kisumu more rapidly. Controlling concentration is therefore a powerful tool for controlling how fast biological, chemical, and industrial processes run.
Lesson 4 Factor 2: Effect of Concentration on Reaction Rate — More Particles, More Collisions	As the concentration of sodium thiosulfate solution decreased (more water was added), the time for the paper cross to disappear increased. The most concentrated solution (C1, full strength) produced the fastest reaction, while the most dilute solution (C4, 25 percent) produced the slowest reaction. When rate was calculated as 1 divided by time and plotted against concentration, the graph showed an approximately directly proportional increase — doubling the concentration roughly doubled the rate.	Higher concentration means more solute particles are present in the same volume of solution. When there are more sodium thiosulfate particles per unit volume, they encounter HCl particles more frequently. More frequent collisions mean more effective collisions per second, so the sulfur precipitate forms faster and the cross disappears sooner. This is the same collision theory used in Lesson 3, but the variable is particle NUMBER rather than particle SPEED.	The concentration effect explains why a more concentrated battery acid corrodes the metal terminals of a matatu faster than a dilute one — more acid particles per unit volume means more frequent collisions with the metal surface, producing corrosion products faster. It also explains why an antacid tablet dissolved in a small amount of water (more concentrated solution) may act faster than one dissolved in a full glass — and why pharmacists must carefully control the concentration of medicines to ensure the correct reaction rate inside the body.
Lesson 5 Lesson 5: Surface Area and Reaction Rate — Grinding, Crushing, and the Secret Inside Solid Reactants	Marble powder reacted with dilute HCl faster than small chips, which reacted faster than large chips — all at the same temperature and concentration. The powder flask showed rapid fizzing and the fastest mass loss. The large-chip flask showed slow, steady bubbling. All three eventually produced the same total mass loss (same amount of reactant used up), but the RATE differed dramatically.	Increasing the surface area of a solid reactant increases the rate of reaction because more particles of the solid are exposed to collisions with the acid particles at any given moment, increasing collision frequency. This is the same collision-frequency principle seen with temperature (Lesson 1) and concentration (Lessons 3–4). Surface area is only relevant for SOLID reactants — dissolved reactants already have maximum particle exposure.	At the particle level, a large marble chip exposes only its outer layer of $CaCO_3$ particles to $HCl(aq)$ particles. When the same mass is ground to powder, millions of new surfaces are created, exposing vastly more $CaCO_3$ particles simultaneously. More exposed particles means more successful collisions per second, so CO_2 and $CaCl_2$ are produced at a higher rate. This explains why grinding maize at the posho mill, crushing charcoal for the jiko, breaking limestone at Bamburi Cement, and chewing githeri thoroughly all use the same principle to speed up chemical or biological reactions

			in everyday Kenyan life.
Lesson 6 Factor 4: Catalysts and Light — Speeding Up Reactions Without Being Used Up	Fresh liver added to hydrogen peroxide produced vigorous bubbling and oxygen gas. Boiled liver produced no detectable reaction. Sand produced no reaction. The same hydrogen peroxide was used in all three tubes.	A catalyst speeds up a reaction without being used up. Enzymes are biological catalysts that work by lowering activation energy. Heating destroys enzyme activity. Iron catalysts are used industrially in fertiliser production. Light energy can drive or accelerate certain reactions.	Catalysts work by providing an alternative reaction pathway with a lower activation energy barrier, meaning a greater fraction of particle collisions have enough energy to be successful — even though the total number of collisions has not changed. This is different from temperature (which increases collision frequency and energy) and from concentration or surface area (which increase collision frequency). Enzymes achieve this through their precise three-dimensional shape, which is destroyed by high temperatures. Light-sensitive reactions are accelerated because light supplies energy directly to reactant particles, increasing the proportion of successful collisions. All these factors — temperature, concentration, surface area, catalysts, and light — ultimately determine whether the particles inside a reaction vessel collide with sufficient energy and correct orientation to form new products, explaining everything from the disappearing thiosulfate cross to the rising froth of uji on a Kisumu stove.
Lesson 7 Light and Catalysts — The Final Rate Factors	In Station A, adding manganese dioxide (MnO_2) to hydrogen peroxide caused immediate vigorous effervescence and relighted a glowing splint, confirming oxygen production — the reaction was dramatically faster than the control with no MnO_2. Crucially, the MnO_2 powder was still visible on the filter paper after the reaction had ended, showing it was not consumed. In Station B, the photosensitive paper exposed to bright light or direct sunlight changed colour significantly faster than the strip kept in dim light, and the strip stored in the dark showed no change at all.	A catalyst is a substance that increases the rate of a chemical reaction by providing an alternative reaction pathway with a lower activation energy, without being consumed in the process. This means that at the same temperature, more collisions have enough energy to result in a successful reaction. Light can also increase the rate of certain chemical reactions (photochemical reactions) by supplying energy directly to reactant particles — similar to the way heat energy increases collision energy in the thiosulfate experiment. Both catalysts and light therefore increase reaction rate through an energy mechanism, but neither changes the concentration or surface area of the reactants.	All five rate factors investigated across Lessons 3–7 (temperature, concentration, surface area, catalysts, and light) increase reaction rate either by increasing the frequency of collisions between particles, by increasing the energy of those collisions, or by lowering the minimum energy (activation energy) that particles need for a collision to be successful. In the anchoring phenomenon, Beaker B (warm, $\sim 50^\circ\text{C}$, simulating Mombasa/Kisumu heat) reacted faster than Beaker A (cold, $\sim 10^\circ\text{C}$, simulating a cool Nairobi highland morning) because heat gave more sodium thiosulfate and HCl particles sufficient energy to collide successfully — the same principle as a catalyst or light providing the energy boost needed to overcome the activation energy barrier. In Kenya, enzymes in the human gut act as biological catalysts to break down

			ugali and githeri efficiently at body temperature; chlorophyll acts as a light-harvesting catalyst in photosynthesis in Rift Valley tea and maize crops; and industrial catalysts are used in fertiliser manufacturing to speed up the Haber process at Kenyan agro-chemical plants.
Lesson 8 Model Building, Assessment, and Community Connection: Closing the Case on Reaction Rates	We observed the disappearing thiosulfate cross for the last time — this time through the lens of all four rate factors. We also observed each groups cumulative particle-collision model and the full set of evidence charts from all seven lessons displayed as a gallery.	We learned that temperature, concentration, surface area, and catalysts all affect the rate of a chemical reaction by changing the frequency or energy of particle collisions — and that this single principle explains phenomena as varied as cooking githeri in Kisumu, drying dagaa at Lake Victoria, rusting of mabati roofs in Mombasa, enzyme activity in the human gut, and industrial reactions at Athi River.	We explained why the cross disappeared faster in the warm beaker on Day 1: higher temperature gave thiosulfate and hydrochloric acid particles more kinetic energy, causing more frequent and more energetic collisions, meaning more collisions exceeded the activation energy and produced sulfur precipitate faster. We also explained the Lake Victoria dagaa scenario and the Factory A vs Factory B antacid scenario using the same collision-theory model — showing that one set of ideas can explain many different Kenyan realities.

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